Appendix A

The Komlós lemma

This appendix proves the form of the Komlós lemma used in Chapter 3, following Delbaen and Schachermayer [1, Lemma A1.1]. The argument is elementary: one maximises the expected value of a bounded concave utility along forward convex combinations and exploits the modulus of concavity to obtain a Cauchy sequence in probability.

Lemma A.1. *Let $(Y_n)_{n\in\mathbb{N}}$ be a sequence of nonnegative random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then there exist forward convex combinations*

$$\widetilde{Y}_n \in \operatorname{conv}\{Y_n, Y_{n+1}, \dots\}, \qquad n \in \mathbb{N},$$

and a measurable function $\widetilde{Y} : \Omega \to [0, \infty]$ such that $\widetilde{Y}_n \to \widetilde{Y}$ $\mathbb{P}$-a.s. as $n \to \infty$. If moreover the set

$$\operatorname{conv}\{Y_n : n \in \mathbb{N}\}$$

is bounded in L^0, then $\widetilde{Y} \in L^0$, i.e. $\widetilde{Y} < \infty$ $\mathbb{P}$-a.s.

Remark A.2. Boundedness in L^0 of a family $\mathcal{A} \subseteq L^0$ means: for every $\varepsilon > 0$ there exists $K \geq 0$ such that

$$\sup_{X\in\mathcal{A}} \mathbb{P}[|X| > K] < \varepsilon.$$

In Chapter 3 the relevant sequence is $(W(\delta_n))_{n\in\mathbb{N}}$ with $\delta_n \in \mathcal{H}$. Hypothesis (A4) bounds the whole gain set $\mathcal{H}_W$ in L^0, and convexity of $\mathcal{H}$ together with convexity of the cost functional yields

$$\sum_{m\geq n} \lambda_{n,m} W(\delta_m) \leq W\left(\sum_{m\geq n} \lambda_{n,m}\delta_m\right)$$

almost surely for every forward convex combination. The right-hand side belongs to $\mathcal{H}_W$, so the forward convex combinations of $(W(\delta_n))_{n\in\mathbb{N}}$ are themselves L^0-bounded and the finiteness conclusion of Lemma A.1 applies.

Proof. Define $u\colon [0,\infty] \to [0,1]$ by $u(x) = 1 - e^{-x}$, with the convention $e^{-\infty} = 0$, so that $u(\infty) = 1$. The function u is continuous, strictly increasing and concave on $[0,\infty]$, and $u(0) = 0$. For each $n \in \mathbb{N}$ set

$$s_n := \sup\big\{\mathbb{E}[u(g)] : g \in \operatorname{conv}\{Y_n, Y_{n+1}, \dots\}\big\} \in [0,1]$$

and choose $g_n \in \operatorname{conv}\{Y_n, Y_{n+1}, \dots\}$ such that

$$\mathbb{E}[u(g_n)] \geq s_n - \frac{1}{n}. \tag{A.1}$$

The sequence $(s_n)_{n\in\mathbb{N}}$ is nonincreasing, hence $s_n \downarrow s_\infty$ for some $s_\infty \in [0,1]$. Combined with (A.1) this yields

$$\lim_{n\to\infty} \mathbb{E}[u(g_n)] = s_\infty.$$

We work with the compact metrizable space $[0,\infty]$. A sequence $(x_k)_{k\in\mathbb{N}}$ in $[0,\infty]$ is Cauchy in this topology if and only if for every $\varepsilon > 0$ there exists k_0 such that for all $k, \ell \geq k_0$,

$$|x_k - x_\ell| < \varepsilon \qquad \text{or} \qquad \min(x_k, x_\ell) > \varepsilon^{-1}.$$

(Here $|x - y|$ is interpreted in the usual way on $[0,\infty)$, with the conventions $|\infty - \infty| = 0$ and $|x - \infty| = \infty$ for $x < \infty$.) The next claim records the modulus of concavity of u on the region where this alternative can fail.

Claim. For every $\varepsilon > 0$ there exists $\beta > 0$ such that, whenever $x, y \in [0,\infty]$ satisfy $|x - y| \geq \varepsilon$ and $\min(x,y) \leq \varepsilon^{-1}$,

$$u\Big(\frac{x+y}{2}\Big) \geq \frac{u(x) + u(y)}{2} + \beta.$$

To see this, assume without loss of generality that $x \leq y$. Then $y \geq x + \varepsilon$ and $x \leq \varepsilon^{-1}$. On $[0,\infty)$ one has the elementary identity

$$u\Big(\frac{x+y}{2}\Big) - \frac{u(x) + u(y)}{2} = \frac{1}{2}\big(e^{-x/2} - e^{-y/2}\big)^2.$$

Hence

$$\begin{aligned}
u\Big(\frac{x+y}{2}\Big) - \frac{u(x) + u(y)}{2} &= \frac{1}{2}\big(e^{-x/2} - e^{-y/2}\big)^2 \\
&\geq \frac{1}{2}\big(e^{-x/2} - e^{-(x+\varepsilon)/2}\big)^2 \\
&= \frac{1}{2}\, e^{-x}\big(1 - e^{-\varepsilon/2}\big)^2 \\
&\geq \frac{1}{2}\, e^{-1/\varepsilon}\big(1 - e^{-\varepsilon/2}\big)^2.
\end{aligned}$$

The right-hand side is a positive constant depending only on ε; call it β. If one of x, y equals ∞, the same lower bound continues to hold by continuity of u at infinity. This proves the claim.

We now show that $(g_n)_{n\in\mathbb{N}}$ is Cauchy in probability with respect to the topology of $[0,\infty]$. Fix $\varepsilon > 0$ and take $\beta > 0$ as in the claim. For $n, m \in \mathbb{N}$, the random variable $(g_n + g_m)/2$ belongs to $\operatorname{conv}\{Y_{n\wedge m}, Y_{n\wedge m+1}, \dots\}$, so

$$\mathbb{E}\Big[u\Big(\frac{g_n + g_m}{2}\Big)\Big] \le s_{n\wedge m}.$$

On the other hand, the claim yields

$$u\Big(\frac{g_n + g_m}{2}\Big) \ge \frac{u(g_n) + u(g_m)}{2} + \beta\, 1_{\{|g_n - g_m| \ge \varepsilon,\, \min(g_n, g_m) \le \varepsilon^{-1}\}},$$

and taking expectations gives

$$\begin{aligned}\beta\,\mathbb{P}\big[|g_n - g_m| \ge \varepsilon,\ \min(g_n, g_m) \le \varepsilon^{-1}\big] &\le \mathbb{E}\Big[u\Big(\frac{g_n + g_m}{2}\Big)\Big] - \frac{1}{2}\big(\mathbb{E}[u(g_n)] + \mathbb{E}[u(g_m)]\big)\\ &\le s_{n\wedge m} - \frac{1}{2}\big(\mathbb{E}[u(g_n)] + \mathbb{E}[u(g_m)]\big).\end{aligned}$$

Using (A.1) and writing $n \le m$ without loss of generality, the right-hand side is at most

$$s_n - \frac{1}{2}\Big(s_n - \frac{1}{n} + s_m - \frac{1}{m}\Big) = \frac{1}{2}(s_n - s_m) + \frac{1}{2n} + \frac{1}{2m}.$$

Since $s_k \downarrow s_\infty$, the right-hand side tends to 0 as $n, m \to \infty$. Therefore

$$\lim_{n,m\to\infty} \mathbb{P}\big[|g_n - g_m| \ge \varepsilon,\ \min(g_n, g_m) \le \varepsilon^{-1}\big] = 0.$$

By the characterisation of the Cauchy property on $[0,\infty]$, the sequence $(g_n)_{n\in\mathbb{N}}$ is Cauchy in probability and thus converges in probability to some measurable $g\colon \Omega \to [0,\infty]$.

Passing to a subsequence $(g_{n_k})_{k\in\mathbb{N}}$ that converges $\mathbb{P}$-a.s. to g, and observing that $n_k \ge k$ for every k (after discarding finitely many terms if necessary), we obtain

$$g_{n_k} \in \operatorname{conv}\{Y_{n_k}, Y_{n_k+1}, \dots\} \subseteq \operatorname{conv}\{Y_k, Y_{k+1}, \dots\}.$$

Setting $\widetilde{Y}_k := g_{n_k}$ and $\widetilde{Y} := g$ yields the first assertion of the lemma.

Finally, suppose $\operatorname{conv}\{Y_n : n \in \mathbb{N}\}$ is bounded in L^0. Then for every $\varepsilon > 0$ there exists $N \ge 0$ such that $\mathbb{P}[h > N] < \varepsilon$ for every $h \in \operatorname{conv}\{Y_n : n \in \mathbb{N}\}$. In particular $\mathbb{P}[\widetilde{Y}_k > N] < \varepsilon$ for every k. On the event $\{\widetilde{Y} > N\}$ one has $\widetilde{Y}_k > N$ for all large k, so

$$1_{\{\widetilde{Y} > N\}} \le \liminf_{k\to\infty} 1_{\{\widetilde{Y}_k > N\}}.$$

Taking expectations and applying Fatou's lemma yields $\mathbb{P}[\widetilde{Y} > N] \le \varepsilon$. Since $\varepsilon > 0$ was arbitrary, $\widetilde{Y} < \infty$ $\mathbb{P}$-a.s., i.e. $\widetilde{Y} \in L^0$. $\square$

Bibliography

[1] F. Delbaen and W. Schachermayer. A general version of the fundamental theorem of asset pricing. *Mathematische Annalen* **300**(1), 463–520, 1994.